EEE407 Tutorial 1

1. A 80 W incandescent light bulb is connected to a 230 V AC supply. After leaving the bulb on for some time, determine:
a) The current flowing in the bulb
b) The resistance of the bulb
Note: In power system, any voltage mentioned is the rms value of the voltage unless otherwise stated.

2. An electric kettle has a resistance of 30 Ω. What is the value of current flowing when it is connected to a 230 V supply? Find also the power rating of the kettle.

3. A street light with resistive element of 130 Ω connected to supply voltage of 230 V is being switched on from 7pm to 7am of the following day. Find the energy consumed by the street light during that period.

4. Calculate daily energy consumed by domestic appliances listed below.
a) 4 units of 1.5 HP air-conditioning unit that operate for 8 hours per day.
b) 32 units of 36 W fluorescence tube lamps that operate for 12 hours per day.
c) 1 unit of 2.0 kW water kettle that operate 15 minutes twice a day.

5. An energy audit has been conducted at a chiller system and recorded approximately 4,606,392 Btu/h of cooling and 250 kW electrical.
a) What is the cooling/heat extraction capacity in Refrigeration Tons (RT).
b) What is the Coefficient of performance (COP) of this system.

6. Use MS Excel to plot the graph and calculate the daily load factor of electricity usage over the 24-hour period as shown in Table 1-1.

Table 1-1

Time	kW	Time	kW	Time	kW	Time	kW
0:00	3688	6:00	1532	12:00	3688	18:00	1818
0:30	3649	6:30	1532	12:30	3792	18:30	1610
1:00	3623	7:00	1519	13:00	3688	19:00	1493
1:30	3987	7:30	1779	13:30	3584	19:30	1520
2:00	4792	8:00	1961	14:00	3155	20:00	1558
2:30	5078	8:30	2610	14:30	3130	20:30	1545
3:00	3545	9:00	3065	15:00	3156	21:00	1545
3:30	3532	9:30	3130	15:30	3311	21:30	1571
4:00	2532	10:00	3027	16:00	3260	22:00	2584
4:30	1532	10:30	3286	16:30	3104	22:30	2467
5:00	1558	11:00	3363	17:00	3013	23:00	3065
5:30	1558	11:30	3597	17:30	2312	23:30	3208

7. **Estimate** the daily load factor of electricity usage over the 24-hour period as shown in Figure 1-1.

a) You may convert the graph in picture format to actual values using http://www.graphreader.com/

b) Calculate the area below the graph using trapezoidal rule for Riemann sums, research online on how to do this with MS Excel.

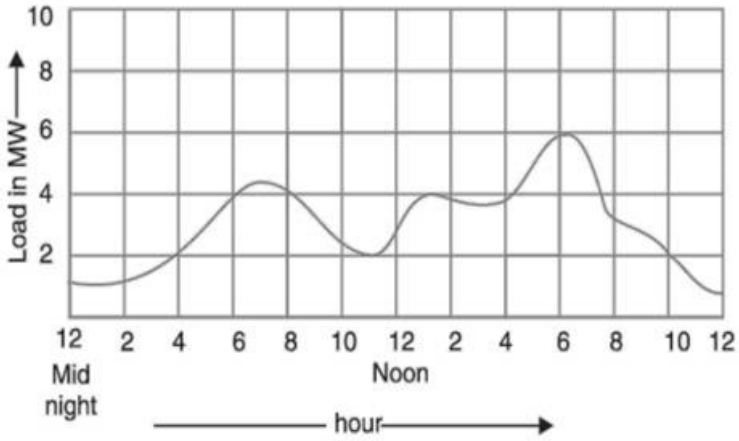


Figure 1-1

8. From question 7, propose a way to improve the daily load factor, considering that the minimum load to keep the office part of building operating is 4MW, what will be the load factor.

9. Based on the Energy efficiency and use of renewable energy for residential buildings - Code of practice (MS 2680:2017), the recommended air conditioning capacity and room size is tabulated in Table 10.

Determine the size/number of air-conditioner for conditions below, show calculation and justify your choice:

a) Living room with an area of 380 ft^2, 3.2 m ceiling height.

b) If there are 5 occupants.

Table 10. Guide on recommended air conditioning capacity and room size

Area to be cooled [ft^2 (m^2)]	Capacity needed [Btu/h (hp)]
180 (17) to 260 (24)	9 000 (1)
270 (25) to 390 (36)	13 500 (1.5)
360 (33) to 520 (48)	18 000 (2)
450 (42) to 650 (60)	22 500 (2.5)
540 (50) to 770 (72)	27 000 (3)
NOTE. An airconditioner of 1 horsepower capacity will remove approximately 9 000 Btu/h of heat.	

Adjustments will have to be made for the following situations:

i) If the room height exceeds 8 ft (2.43 m) the capacities will have to be increased by 10 % for every 1 ft (0.3 m) increase in height.

ii) If there are more than two persons who will regularly occupy the room, add 500 Btu/h for each additional person.

10. Define British thermal unit and refrigeration ton.

11. What are the general procedures for establishing an energy management program?

12. List down 3 examples of zero/low, medium, and high-cost energy conservation measures (ECM).

13. How does electrical energy utilization affect the environment and what are the alternatives to reduce its impact?

14. What are the different methods for measuring electrical energy utilization in households and industries?

15. What are the different technologies for renewable energy production and how can they be utilized to reduce the dependence on traditional electrical energy sources?

16. How can energy storage technologies be utilized to improve electrical energy utilization and reduce energy waste?

17. What are the different approaches for improving electrical energy efficiency in households and industries, and what are their impacts?